

Thanmay M Shetty

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Summary

Agentic AI Engineer with experience designing and deploying LLM-powered systems, AI agents, and production-grade RAG pipelines across healthcare, agri-tech, scientific data, and NLP domains. Designed RAG architectures and AI agent pipelines from scratch, improving query accuracy by 20% and model accuracy from 76% to 94% through fine-tuning. Top-3 finisher across 7 national and inter-collegiate hackathons; managed a 15-member technical chapter with a ₹50,000 annual budget.

Education

B.E. in Artificial Intelligence & Machine Learning	2027
BMS Institute of Technology and Management (BMSITM), Bengaluru	CGPA: 9.01
PUC in PCMC	2023
Christ Academy Junior College, Bengaluru	95%

Technical Skills

Languages: Python, JavaScript, C, Java

Agentic AI & LLMs: AI Agents, Agentic Workflows, RAG Pipelines, Prompt Engineering, Tool Calling, Memory Systems, LLM Evaluation

Frameworks: LangChain, LangGraph, Langflow, OpenAI APIs, Gemini, Claude, Ollama, FastAPI, Flask

Backend & APIs: FastAPI, Flask, REST APIs, OpenAPI, API Design, Versioning, Request/Response Schemas, PyTorch, TensorFlow

Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, SQL, relational databases, data modelling, NETCDF/ARGO workflows

Infrastructure: Docker, CI/CD, Containerised Deployment, Git/GitHub, API Testing

Algorithms & DS: Data structures, algorithm design, complexity analysis

Project Experience

BioBloom Ecosystem — 1st Place, RNSIT National Hackathon

- Architected and deployed multilingual agentic RAG platform (AgriGain) using LangChain workflows, tool-enabled agents, retrieval pipelines, memory-driven interactions, and FastAPI backend services. guards, integrated cloud database, backend, and frontend; deployed to Vercel.
- Improved agent response quality through iterative prompt engineering, structured output validation, tool orchestration, and evaluation-driven experimentation, achieving 90% structured response success rate; delivered results in English, Hindi, and Kannada.
- Curated 3GB of crop rotation research papers (AgriKami) as knowledge base for a RAG-based crop planning system.

ARGO NETCDF Explorer — Scientific Data RAG Pipeline

- Constructed RAG pipeline from scratch over 5GB of oceanographic data; parsed, chunked, and embedded ~20,000 profile chunks with structured metadata into a vector database with filtered retrieval.
- Improved natural-language query accuracy from ~70% to ~90% across 5 prompt engineering iterations; built interactive dashboard with 6–7 data widgets including location maps and multi-year trend graphs.

Sentiment Analysis API — DistilBERT

- Fine-tuned DistilBERT on 3.6M Amazon review samples, improving accuracy from 76% (base model) to 94% after fine-tuning; deployed scalable REST API with ~500ms average inference latency.
- Engineered modular ML and evaluation pipeline with automated testing, validation workflows, CI/CD deployment, and reproducible experimentation for model benchmarking.

Quantum-AI Cognitive Health System — Solo Exploration

- Developed full-stack AI platform integrating real-time inference pipelines, multimodal processing, recommendation workflows, and API-driven agent interactions across 24 endpoints. API endpoints; classified 3 cognitive states and matched task difficulty from a pool of 5 learning recommendations.
- Processed 157,000+ down sampled EEG time points across 24 upload sessions; implemented QUBO-based task optimizer with classical simulated annealing fallback and offline quantum-classical VQC pipeline using PennyLane and PyTorch.
- Deployed 2-service Docker Compose setup with 6 live dashboard components, JWT authentication, EDF/CSV upload support, and PDF export for clinical reporting.

Smart Attendance System — Computer Vision

- Implemented real-time face recognition system processing 60 FPS, achieving ~1–2 second recognition latency and 5-second registration flow for classroom environments.

Additional GitHub Work

github.com/FrostyZ07

- Developed AI agents, RAG systems, automation scripts, internal tooling, evaluation pipelines, and full-stack AI applications using modern LLM frameworks and developer workflows.

Certifications

- Introduction to LLM — NPTEL
- Leadership and Team Effectiveness — Scored in top 5% nationally

Achievements

- 1st Place — RNSIT National Hackathon, 24-hour AI/ML hackathon (2025)
- 3rd Place — KSEEM National Level Hackathon (2025)
- 2nd Place — Build and Conquer Inter-Collegiate Hackathon (2024)
- Ranked in top 3 across 7 hackathons spanning Agri-tech, health-tech, and full-stack AI domains.

Leadership & Other

Treasurer — IEEE Computational Intelligence Society (CIS)

2024 – 2025

Managed ₹45,000–50,000 annual budget for a 15-member IEEE CIS chapter across 6 events: 3 technical workshops (50–70 attendees each), an inter-collegiate hackathon/ideathon with 50+ teams, and a Winter of Projects mentoring 4 juniors on AI/ML; served as technical spokesperson for LLM and RAG system demos to non-technical stakeholders.